Comprehensive Exam Study Guide for ADT308

Table of Contents

Exam Overview

Course Code: ADT308
Course Name: Comprehensive Course Work
Maximum Marks: 50
Duration: 1 Hour
Question Format: 50 Multiple Choice Questions (1 mark each)
Negative Marking: None
Subjects Covered:

- Introduction to Machine Learning
- Data Structures
- Operating Systems
- Database Management Systems
- Foundations of Data Science

Pattern Analysis of Question Papers

Based on the analysis of both the 2023 question paper and the model question paper, the following patterns were identified:

Similarities

1. **Format and Structure:**
    - Both contain exactly 50 multiple-choice questions
    - Same duration (1 hour) and marking scheme
    - No negative marking in either paper

2. **Subject Coverage:**
   - Both cover the same five core subjects with similar weightage
   - Machine Learning slightly emphasized in both papers
3. **Identical Questions:**
   - Several questions appear in both papers with identical or very similar wording
   - Key concepts like PCA, stack operations, memory management, and ER diagrams are repeatedly tested
4. **Difficulty Level:**
   - Both focus on fundamental concepts rather than complex applications
   - Understanding of basic principles is emphasized over calculations

Differences

1. **Question Ordering:**
   - The 2023 paper groups questions by subject area
   - The model paper has a more mixed arrangement of topics
2. **Minor Variations in Topic Focus:**
   - Slight differences in emphasis within each subject area

Topic Distribution and Weightage

Based on question frequency analysis:

Subject	Weightage	Number of Questions
Machine Learning	22%	11 questions
Data Structures	20%	10 questions
Operating Systems	20%	10 questions
Database Management Systems	20%	10 questions
Foundations of Data Science	18%	9 questions

Subject-wise Important Topics

1. Machine Learning (High Priority)

Topic	Importance	Question Frequency
Supervised vs Unsupervised Learning	Very High	5+ questions
Dimension Reduction & PCA	High	2-3 questions
Support Vector Machines	Medium	1-2 questions
Clustering Algorithms	Medium	1-2 questions
Model Performance Metrics	High	2-3 questions
Feature Selection Methods	Medium	1-2 questions

Key Concepts to Master:

- Difference between supervised, unsupervised, and reinforcement learning
- Classification vs regression problems
- Metrics: accuracy, precision, F1 score
- PCA for dimension reduction
- Forward and backward feature selection
- K-means clustering and Lloyd's algorithm
- Random Forest as an ensemble method

Key Formulas:

- Accuracy = (TP + TN) / (TP + TN + FP + FN)
- Precision = TP / (TP + FP)
- F1 Score = 2 * (Precision * Recall) / (Precision + Recall)

2. Data Structures (High Priority)

Topic	Importance	Question Frequency
Stack and Queue Operations	Very High	3-4 questions
Tree Structures	High	3-4 questions
Searching & Sorting	High	2-3 questions
Time Complexity Analysis	Medium	1-2 questions
Binary Search Trees	High	2-3 questions

Key Concepts to Master:

- Stack operations (push, pop)
- Queue operations (enqueue, dequeue)
- Binary search tree properties and operations
- In-order, pre-order, post-order traversals

- Time complexity of common algorithms
- Sorting algorithms (quick sort, merge sort, insertion sort)
- Binary search algorithm

Example Problems:

- Calculating final values after sequence of stack/queue operations
- Identifying appropriate data structures for specific operations
- Converting infix to postfix expressions
- Determining time complexity of algorithms

3. Operating Systems (Medium-High Priority)

Topic	Importance	Question Frequency
Memory Management & Paging	Very High	3-4 questions
Process Scheduling	High	2-3 questions
Deadlock Prevention	Medium	2 questions
File Systems	Medium	1-2 questions
Process States	Medium	1-2 questions

Key Concepts to Master:

- Page size calculations and internal fragmentation
- CPU scheduling algorithms (especially Round Robin)
- Deadlock conditions and prevention
- Process states and transitions
- Banker's algorithm
- Memory address translation

Key Calculations:

- Internal fragmentation = Page size - (Process size % Page size)
- Addressable memory calculation with page table entries
- CPU utilization percentage

4. Database Management Systems (Medium-High Priority)

Topic	Importance	Question Frequency
ER Diagrams & Relational Models	High	3 questions
SQL Commands	Medium	2-3 questions
Normalization	High	2-3 questions
Keys (Primary, Foreign)	Medium	2 questions
Functional Dependencies	Medium	2 questions

Key Concepts to Master:

- Converting ER diagrams to relational models
- Normal forms (1NF through BCNF)
- Functional dependencies and their implications
- SQL operations (SELECT, JOIN, etc.)
- Primary key and foreign key constraints
- Entity relationships (one-to-many, many-to-many)

Important Rules:

- For many-to-many relationships, create a separate relation
- Normalization criteria for each normal form
- Rules for identifying candidate keys

5. Foundation of Data Science (Medium Priority)

Topic	Importance	Question Frequency
Regression Analysis	High	3 questions
Data Analytics Methods	Medium	2 questions
Hypothesis Testing	Medium	1-2 questions
Data Visualization	Medium	1-2 questions
Quantitative Analysis	Medium	2 questions

Key Concepts to Master:

- Linear regression principles
- Types of linear regression (simple vs multiple)
- Data visualization techniques for different data types
- Hypothesis testing fundamentals
- Data quantification methods
- Statistical methods for data analysis

Visualization Guidelines:

- Scatterplots for relationship between two quantitative variables
- Bar graphs for categorical comparisons
- Line graphs for time series data

Repeating Questions Analysis

The following questions appeared in both papers with identical or very similar wording:

Machine Learning

1. **PCA Purpose:** Principal Components Analysis (PCA) is used for dimension reduction
2. **Supervised vs. Unsupervised Learning:** Definition and examples
3. **Classification vs. Regression:** Types of supervised learning
4. **Machine Learning Model Output:** The output of training process is a machine learning model
5. **Random Forest:** Ensemble learning method based on bagging

Data Structures

1. **Stack and Queue Operations:** Calculating values after a sequence of operations
2. **Postfix Expression:** Converting infix to postfix expressions
3. **Binary Search Tree:** Properties and traversal methods
4. **Time Complexity:** Binary search algorithm complexity (O(log2n))
5. **Merge Sort:** As an example of divide and conquer algorithm

Operating Systems

1. **Internal Fragmentation:** Calculations with page size and process size
2. **Memory Addressing:** Addressable memory with page table entries
3. **Round Robin Scheduling:** Characteristics and uses
4. **Deadlock Prevention:** Methods and conditions
5. **Process States:** Transitions between states in timeshare systems

Database Management Systems

1. **ER Diagram to Tables:** Minimum number of tables needed for given relationships
2. **SQL Commands:** DROP vs DELETE operations
3. **Normal Forms:** Determining highest normal form of a relation
4. **Entity Relationships:** Cardinality in relationships
5. **Primary and Foreign Keys:** Identification and constraints

Foundation of Data Science

1. **Linear Regression:** Finding best fit line between variables
2. **Data Visualization:** Using scatterplots for quantitative variables
3. **Quantification of Data:** Process referred to as coding or enumeration
4. **Regression Analysis:** For modeling relationships in data
5. **Data Science Applications:** Modern conception and development

Strategic Preparation Plan

1. Two-Week Plan

Week 1: Foundation Building

- Day 1-2: Machine Learning basics and algorithms
- Day 3-4: Data Structures fundamentals
- Day 5-6: Operating Systems core concepts
- Day 7: Database Management Systems basics

Week 2: Advanced Review and Practice

- Day 8-9: Data Science concepts and applications
- Day 10-11: Practice with previous question papers
- Day 12-13: Focus on repeating question patterns
- Day 14: Final revision of key concepts

2. Study Techniques

- **Concept Mapping:** Create visual connections between related concepts
- **Active Recall:** Test yourself frequently on key concepts
- **Spaced Repetition:** Review material at increasing intervals
- **Practice MCQs:** Simulate exam conditions with timed practice
- **Problem Variations:** Create variations of calculation problems

3. Exam Day Strategy

- **Time Management:** 72 seconds per question on average
- **Answer Selection:** Eliminate obviously wrong options first
- **Question Prioritization:** Answer familiar questions first
- **Review Strategy:** Mark uncertain questions for review if time permits
- **Guessing Strategy:** With no negative marking, always select an option

Quick Reference Guide

Top 10 Most Important Topics

1. **Supervised vs. Unsupervised Learning** (Machine Learning)
2. **Stack and Queue Operations** (Data Structures)
3. **Memory Management & Paging** (Operating Systems)
4. **ER Diagrams to Relational Models** (DBMS)
5. **Linear Regression Concepts** (Data Science)
6. **Binary Search Trees** (Data Structures)
7. **Process Scheduling Algorithms** (Operating Systems)
8. **Normalization Forms** (DBMS)
9. **Principal Component Analysis** (Machine Learning)
10. **Data Visualization Techniques** (Data Science)

Common Calculation Questions

1. **Internal Fragmentation Calculation:**
   - If page size is 4KB and process size is 103KB, internal fragmentation = 1KB
   - Formula: Internal Fragmentation = Page Size - (Process Size % Page Size)
2. **Stack and Queue Operations:**
   - Follow the operations step by step tracking values
   - Pay attention to the order of operations
3. **Addressable Memory with Paging:**
   - With frame size 4KB and page table entry of 2 bytes, addressable memory = 2^28 bytes
   - Formula: Addressable Memory = (2^(8*page_entry_size)) * frame_size

Practice Questions

Machine Learning

1. **Question:** The problem of finding hidden structure in unlabelled data is called **Options:** a) Supervised Learning b) Unsupervised Learning c) Reinforcement Learning d) None **Answer:** b) Unsupervised Learning
2. **Question:** The ratio of correctly predicted labels and the total number of predicted labels is termed as **Options:** a) Accuracy b) Precision c) F1 score d) Confusion matrix **Answer:** b) Precision

Data Structures

1. **Question:** Consider the following sequence of operations on an empty stack: push(22); push(43); pop(); push(55); push(12); s=pop(); Consider the following sequence of operations on an empty queue: enqueue(32);enqueue(27); dequeue(); enqueue(38); enqueue(12); q=dequeue(); The value of s+q is **Options:** a) 44 b) 54 c) 39 d) 70 **Answer:** c) 39
2. **Question:** Select the postfix expression for the infix expression a+b-c+d*(e/f). **Options:** a) ab+c-d+e*f/ b) ab+c-def/*+ c) abc-+def/+ *d) ab+c-def/*+ **Answer:** d) ab+c-def/*+

Operating Systems

1. **Question:** Calculate the internal fragmentation if page size is 4KB and process size is 103KB. **Options:** a) 1KB b) 2KB c) 3KB d) 4KB **Answer:** a) 1KB
2. **Question:** If frame size is 4KB then a paging system with page table entry of 2 bytes can address _______ bytes of physical memory. **Options:** a) 2^12 b) 2^16 c) 2^18 d) 2^28 **Answer:** d) 2^28

Database Management Systems

1. **Question:** Let E1, E2 and E3 be three entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many, R2 is many-to-many. R3 is another relationship between E2 and E3 which is many-to-many. R1, R2 and R3 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model? **Options:** a) 3 b) 4 c) 5 d) 6 **Answer:** c) 5
2. **Question:** Find the highest normal form of a relation R(A,B,C,D,E) with Functional Dependency {B->A, A->C, BC->D, AC->BE) **Options:** a) 1NF b) 2NF c) 3NF d) BCNF **Answer:** c) 3NF

Foundations of Data Science

1. **Question:** Linear Regression is the supervised machine learning model in which the model finds the best fit _______ between the independent and dependent variable. **Options:** a) Linear line b) Non linear line c) Curved line d) All of the above **Answer:** a) Linear line
2. **Question:** _______ are used when we want to visually examine the relationship between two quantitative variables. **Options:** a) Scatterplot b) Bar graph c) Pie chart d) Line graph **Answer:** a) Scatterplot

---

This study guide was created based on the analysis of previous question papers and is designed to help you prepare effectively for the ADT308 Comprehensive Course Work examination.